

Scalar-Carrier CP4 Geometry Robustness Release for HAOS-IIP

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Abstract

This paper freezes release 51.3 of HAOS-IIP as the numbered robustness follow-up to the already-positive scalar-carrier CP4 geometry bridge. The earlier scalar bridge had shown that one common local scalar kernel-graph family can organize Green response, heat behavior, shell-arrival structure, and first-shell low-mode organization under one common coarse Euclidean shell reconstruction. Release 51.3 asks the next honest question: does that bridge survive bounded stress on the same carrier? The answer is yes on the tested window. Across all tested mild point-set disorder cases and bounded kernel-family variants, the same scalar carrier remains compatible with the same 3D-like geometry read, while the extracted coordinate-response tensor stays close to isotropic. The result strengthens the repository’s geometry authority boundary from a one-family bridge statement to a bounded robust-carrier statement, without promoting the result into broad universality, curvature, or ontology.

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1 Scope

Release 51.3 is a narrow scalar-carrier synthesis freeze.

It does not add:

- a new vector validation claim beyond the already frozen 51.1 and 51.2 line;
- a broader continuum or spacetime claim than the bounded scalar carrier actually supports;
- broad universality across arbitrary kernels, disorder classes, or irregular substrates;
- curvature extraction, conserved currents, or geometry-as-ontology language.

It does add:

- a numbered release for the first bounded robustness pass on the scalar-carrier CP4 geometry bridge;
- one compact authority statement covering mild point-set disorder, bounded kernel-family variation, and coordinate-response extraction on the same carrier;
- a sharper boundary between what now counts as a robust scalar geometry read and what remains open.

2 Where the Repository Stood Before 51.3

Before this release, the scalar kernel-graph line had already crossed one important threshold: the same common scalar carrier had been used to organize

- Green response;
- heat shell behavior;
- shell-arrival slopes;
- first-shell low-mode organization.

That resolved the earlier preflight mismatch in which Green response and late-phase distance-surrogate diagnostics had been living on different substrate families. But the resulting authority boundary was still narrow:

the scalar-carrier geometry bridge is positive on one tested local cubic family.

Release 51.3 exists to test whether that statement survives the first bounded stress program on the same carrier rather than being left as a one-family coincidence.

3 The 51.3 Robustness Program

The same scalar carrier is kept fixed:

- scalar kernel graph on $3D$ cubic point clouds;
- local regime $c_\epsilon = 0.5$;
- same Green / heat / shell-arrival / first-shell low-mode observables;
- same Euclidean shell reconstruction centered at the source.

The bounded robustness program then adds three controlled extensions.

3.1 Mild point-set disorder

The first pass perturbs the point cloud itself while preserving the same shell reconstruction logic.

- sizes: $n = 11, 13$;
- jitter fractions: 0.00, 0.02, 0.04;
- kernel family fixed to `gaussian_local`.

For jittered cases, shell bins are assigned relative to the corresponding clean-grid shell at the same n , so the coarse reconstruction is not silently changed at the same time as the point set.

3.2 Bounded kernel-family variation

The second pass changes the local kernel law while preserving the same operator role and the same scalar carrier.

- sizes: $n = 11, 13$;
- families: `gaussian_local`, `gaussian_half`, `inverse_quadratic`.

This is not a universality proof. It is only the first bounded family test.

3.3 Coordinate-response extraction

The third pass asks whether the same carrier still supports a metric-like response read after bounded variation. On every case the coordinate-response stiffness tensor is extracted on the orthonormalized coordinate basis:

$$K_{\text{coord}} = Q^T \left(-\hat{L}_h \right) Q.$$

If the same geometry read remains valid on the tested carrier, this tensor should stay close to a scalar multiple of the identity.

4 Frozen Quantitative Summary

Table 1 records the compact bounded result frozen by 51.3.

The key point is not just that the best cases remain positive. The weakest passing cases still remain comfortably inside the bounded window.

5 What 51.3 Now Supports

The strongest honest scalar-geometry statement now supported by the repository is:

the scalar-carrier CP4 geometry closure remains stable under the first bounded robustness window of mild point-set disorder, bounded kernel-family variation, and coordinate-response extraction on the tested carrier.

That is stronger than the earlier bridge statement because it promotes the scalar carrier from a one-family positive example to a bounded robust carrier.

This does not replace the original bridge. It makes that bridge harder to dismiss as an artifact of one exact clean cubic setup.

Channel	Status	Frozen read
Mild disorder pass	pass	All 6/6 tested disorder cases pass the shared geometry thresholds.
Kernel-family pass	pass	All 6/6 tested kernel-family cases pass the shared geometry thresholds.
Metric anisotropy	bounded	Maximum anisotropy across all tested cases is 6.77×10^{-4} .
Metric off-diagonal ratio	bounded	Maximum off-diagonal ratio across all tested cases is 1.42×10^{-3} .
Weakest disorder case	pass	<code>disorder n=11 jitter=0.000</code> with shell-arrival fit $R^2 = 0.9797$.
Weakest kernel case	pass	<code>kernel n=13 family=gaussian_half</code> with shell-arrival fit $R^2 = 0.9732$.

Table 1: Bounded scalar-carrier robustness summary frozen by release 51.3.

6 What 51.3 Still Does Not Say

Release 51.3 still does not justify the following stronger claims:

- broad universality across large kernel families;
- closure on strong disorder, irregular sampling, or arbitrary substrate classes;
- curvature extraction from the same carrier;
- conserved-current recovery;
- a repo-wide geometry closure claim across all lines of evidence;
- spacetime or ontology language.

Those remain later questions.

7 Practical Consequence

The scalar carrier is now the repository’s strongest bounded geometry anchor.

It already had:

- the strongest bounded scalar continuum-facing control;
- one common-family geometry bridge for Green response, heat behavior, shell-arrival structure, and low-mode organization.

Release 51.3 adds:

- bounded mild-disorder robustness;
- bounded kernel-family robustness;
- bounded coordinate-response / metric extraction robustness.

So the next honest scalar-geometry step is no longer “does any geometry-like read survive at all?” It is one of the following stronger programs:

- wider universality on the same scalar carrier;
- stronger response / metric extraction beyond the coordinate stiffness tensor;
- careful curvature-like or response-like closure on the same carrier.

8 Conclusion

Release 51.3 freezes the bounded robustness follow-up to the scalar-carrier CP4 geometry bridge.

The result is positive on the tested window. Mild point-set disorder, bounded kernel-family variation, and coordinate-response extraction all remain compatible with the same 3D-like geometry read on the same scalar carrier. That upgrades the scalar geometry line from a one-family bridge result to a bounded robust-carrier statement.

That is the right stopping point for this release: stronger than the initial bridge, but still disciplined about what the repository has not yet earned.